Nanoscale Grating Reconstruction via GIXRF Optimization

Rigorous Formulation, Parameter Sensitivity & Fixation, and Beam Divergence Modeling

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Abstract

This document provides a self-contained, mathematically rigorous reference for the reconstruction of periodic nanostructured surfaces from grazing-incidence X-ray fluorescence (GIXRF) measurements. Focusing on the benchmark experiment from Anna Andrle's MS thesis (Configuration B: $Si_3N_4/SiO_2/Si$ periodic line grating at $E = 680$ eV), we detail: (1) the forward vectorial electromagnetic solver and semi-analytic modal intensity integration, (2) the objective loss function and analytical scale factor elimination, (3) the physical origin, mathematical formulation, and discrete implementation of X-ray beam divergence via Gaussian convolution, (4) a comprehensive sensitivity analysis justifying the fixation of insensitive parameters (t_{sub}, pitch, etch offset, roughness), and (5) the non-linear least-squares optimization mechanics with parameter covariance and confidence interval estimation.

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1 Introduction and Physical System

Grazing-incidence X-ray fluorescence (GIXRF) combines X-ray standing wave (XSW) field effects with element-specific characteristic fluorescence emission. When a periodic grating is illuminated at grazing angles of incidence ($\alpha \leq 15°$), interference between the incident and diffracted waves establishes a nanoscale electromagnetic standing wave field inside and around the grating lines. Because characteristic X-ray fluorescence emission is directly proportional to the local electric field intensity at the position of the emitting atoms, scanning the grazing angle α provides a sensitive probe of nanoscale shape, dimensions, and composition.

1.1 Configuration B Benchmark ($E = 680$ eV)

The target system is Anna Andrle's Configuration B, experimentally measured at the BESSY II plane-grating monochromator (PGM) beamline:

- **Sample**: Periodic silicon nitride (Si_3N_4) trapezoidal line grating on a silicon (Si) substrate, covered by a native/thermal silicon dioxide (SiO_2) surface oxide cap and groove oxide.
- **Nominal Pitch**: $p = 100.834$ nm along the periodic x-axis. The grating is translation invariant along y.
- **Excitation**: Photon energy $E = 680.0$ eV ($\lambda = 1.8233$ nm), in p-polarization, with azimuthal angle $\phi = 0°$ (incident beam aligned parallel to the grating lines).
- **Dual Fluorescence Channels**: At $E = 680$ eV, the photon energy is above both the Nitrogen K-edge (409.9 eV) and the Oxygen K-edge (543.1 eV), exciting:

1. N-Kα emission ($E_{\text{emit}} = 392.4$ eV), originating exclusively from the Si_3N_4 trapezoidal core.
2. O-Kα emission ($E_{\text{emit}} = 524.9$ eV), originating from the SiO_2 top cap and trench groove oxide.

1.2 Multislice Spatial Discretization

The continuous trapezoidal profile is discretized into N horizontal slices (z-layers). In each layer j of thickness Δz_j:

$$w_{\text{N}}(z) = w + \left(z - \frac{h}{2}\right) \frac{2}{\tan(\text{swa})}, \tag{1}$$

where h is the line height, w is the middle line width (critical dimension at half height $z = h/2$), and swa is the sidewall angle. Each slice is composed of discrete piecewise-constant material boxes: a central Si_3N_4 box flanked by SiO_2 sidewall boxes, with groove oxide filling the trench up to thickness t_g.

2 Vectorial Forward Solver and Observable Integration

2.1 Coupled Maxwell Eigenproblem

The solver implements the rigorous coupled-wave analysis (RCWA) for non-magnetic media ($\varepsilon(x, z) = 1 + \chi(x, z)$). Eliminating longitudinal field components yields the coupled first-order system for tangential electric and magnetic Fourier harmonics:

$$\frac{\mathrm{d}\mathbf{e}_t}{\mathrm{d}z} = \mathrm{i}k_0 P \mathbf{h}_t, \qquad \frac{\mathrm{d}\mathbf{h}_t}{\mathrm{d}z} = \mathrm{i}k_0 Q \mathbf{e}_t, \tag{2}$$

leading to the quadratic eigenvalue problem:

$$\left(k_0^2 PQ - k_{z,n}^2 I\right) \mathbf{w}_n = 0, \qquad q_n = k_{z,n}/k_0, \tag{3}$$

where P and Q incorporate Li's inverse Fourier factorization $A = [1/\varepsilon]^{-1}$ to satisfy normal displacement continuity across lateral material boundaries.

2.2 Semi-Analytic Electric Intensity Modal Integration

For isotropic photoabsorption, the local fluorescence excitation is governed by the total electric intensity:

$$|\mathbf{E}(\mathbf{r})|^2 = |E_x(\mathbf{r})|^2 + |E_y(\mathbf{r})|^2 + |E_z(\mathbf{r})|^2. \tag{4}$$

The fluorescence yield emitted by material m within layer j of thickness d is computed directly from modal coefficients without sampling a real-space spatial grid:

$$Y_{j,m} = \sum_{a,b} c_a c_b^* L_{ab,m} \widetilde{U}_{ab}, \tag{5}$$

where c_a, c_b are boundary-anchored modal amplitudes, $L_{ab,m}$ are exact analytical lateral Fourier overlaps over the material box intervals:

$$L_{ab,m} = \int_{\text{boxes}} \mathbf{u}_a(x) \cdot \mathbf{u}_b^*(x)\, \mathrm{d}x = \sum_{\text{box}} w_{\text{box}}\, \text{sinc}\left(\frac{\Delta k_x w_{\text{box}}}{2\pi}\right) e^{\mathrm{i}\Delta k_x x_{\text{mid}}} (\mathbf{v}_a \cdot \mathbf{v}_b^*), \tag{6}$$

and $\widetilde{U}_{ab}$ are bounded, stable depth overlap integrals incorporating the linear self-attenuation coefficient μ_m:

$$\widetilde{U}_{ab} = \int_{z_{\text{top}}}^{z_{\text{bottom}}} e^{\mathrm{i}\kappa_a (z - z_a) - \mathrm{i}\kappa_b^* (z - z_b)} e^{-\mu_m z}\, \mathrm{d}z. \tag{7}$$

For $E = 680$ eV excitation, calibrated linear escape attenuation coefficients are: $\mu_{\text{N}} = 2.75 \times 10^{-3}$ nm^{-1} and $\mu_{\text{O}} = 2.33 \times 10^{-3}$ nm^{-1}.

3 Beam Divergence: Gaussian Filter and Curve Convolution

3.1 Physical Origin

In synchrotron beamline experiments (e.g., BESSY II PGM), the incident beam delivered by the monochromator mirror system is not an idealized monochromatic plane wave with a single mathematical wavevector. Instead, the incident beam exhibits a finite angular distribution characterized by an angular beam divergence $\sigma_{\text{div}} \leq 0.01^\circ$ (≈ 0.175 mrad).

At grazing incidence, features in the fluorescence yield curve—such as the sharp critical-angle surge and high-frequency standing wave interference fringes—can be smoothed by this angular spread. Neglecting beam divergence introduces systematic errors in the reconstructed sidewall angle and oxide layer thicknesses.

3.2 Continuous 1D Convolution Formulation

Let $I_k(\alpha)$ denote the theoretical fluorescence yield calculated for element $k \in \{\text{N}, \text{O}\}$ as a function of the central grazing incidence angle α. The observed intensity is the continuous convolution of $I_k(\alpha)$ with the 1D Gaussian divergence profile $G_{\sigma_{\text{div}}}$:

$$I_{\text{conv},k}(\alpha) = (I_k * G_{\sigma_{\text{div}}})(\alpha) = \int_{-\infty}^{\infty} I_k(\alpha')\, G_{\sigma_{\text{div}}}(\alpha - \alpha')\, \mathrm{d}\alpha', \tag{8}$$

where the normalized Gaussian kernel is defined as:

$$G_{\sigma_{\text{div}}}(\Delta\alpha) = \frac{1}{\sqrt{2\pi}\sigma_{\text{div}}} \exp\left(-\frac{(\Delta\alpha)^2}{2\sigma_{\text{div}}^2}\right), \qquad \int_{-\infty}^{\infty} G_{\sigma_{\text{div}}}(\Delta\alpha)\, \mathrm{d}(\Delta\alpha) = 1. \tag{9}$$

3.3 Discrete Realization on Non-Uniform Angular Grids

In experimental data acquisition, angular grids are frequently non-uniform. In the reference dataset (`20200417_BestFit.npy`), the 180 measured angles span $\alpha \in [0.50075^\circ, 14.90075^\circ]$ with two distinct step regimes:

$$\Delta\alpha = \begin{cases} 0.05^\circ, & \alpha < 4.0^\circ \quad \text{(high-resolution standing wave regime)}, \\ 0.10^\circ, & \alpha \geq 4.0^\circ \quad \text{(high-angle bulk regime)}. \end{cases} \tag{10}$$

Standard fast Fourier transform (FFT) convolution filters (such as `scipy.ndimage.gaussian_filter1d`) assume strictly uniform grid spacing and fail or introduce distortive interpolation artifacts on non-uniform data.

To guarantee mathematical rigor, flux conservation, and seamless gradient computation, we construct a discrete normalized convolution operator matrix $\widetilde{M} \in \mathbb{R}^{N \times N}$.

3.3.1 Quadrature Weighting and Matrix Construction

Each grid point α_j is assigned a trapezoidal quadrature integration weight $\Delta\alpha_j$:

$$\Delta\alpha_0 = \frac{\alpha_1 - \alpha_0}{2}, \quad \Delta\alpha_{N-1} = \frac{\alpha_{N-1} - \alpha_{N-2}}{2}, \quad \Delta\alpha_j = \frac{\alpha_{j+1} - \alpha_{j-1}}{2} \quad (0 < j < N-1). \tag{11}$$

The unnormalized kernel matrix elements are:

$$M_{ij} = \exp\left(-\frac{(\alpha_i - \alpha_j)^2}{2\sigma_{\text{div}}^2}\right) \cdot \Delta\alpha_j. \tag{12}$$

Each row i is subsequently normalized by its row sum:

$$\widetilde{M}_{ij} = \frac{M_{ij}}{\sum_{m=1}^{N} M_{im}}. \tag{13}$$

The convolved fluorescence curve is then evaluated via a single dense matrix-vector product:

$$\mathbf{I}_{\mathrm{conv},k} = \widetilde{M}\,\mathbf{I}_{\mathrm{calc},k}. \tag{14}$$

3.3.2 Key Mathematical Properties

1. **Exact Flux Conservation**: $\sum_i I_{\mathrm{conv}}(\alpha_i)\Delta\alpha_i = \sum_j I_{\mathrm{calc}}(\alpha_j)\Delta\alpha_j$ is satisfied to machine precision.
2. **Identity Limit**: In the limit $\sigma_{\mathrm{div}} \to 0$, $\widetilde{M} \to I_{N\times N}$, recovering the unconvolved solution.
3. **Constant Signal Invariance**: For any constant baseline I_0, $\widetilde{M} \cdot (I_0\mathbf{1}) = I_0\mathbf{1}$, eliminating edge drop-offs.
4. **Computational Efficiency**: The $N \times N$ matrix $\widetilde{M}$ depends purely on the static angle array $\boldsymbol{\alpha}$ and σ_{div}, and is precomputed once in $O(N^2)$ operations (~ 10 μs execution time).

4 Optimization Algorithm and Mathematical Formulation

4.1 Weighted Chi-Square Objective Function

The reconstruction problem is formulated as a simultaneous non-linear weighted least-squares minimization across all measured angles for both fluorescent elements:

$$\chi^2(\mathbf{p}) = \sum_{k\in\{\mathrm{N,O}\}} \sum_{i=1}^{M_k} \left(\frac{I_{\mathrm{exp},k}(\alpha_i) - c_k I_{\mathrm{conv},k}(\alpha_i;\mathbf{p})}{\sigma_{k,i}} \right)^2, \tag{15}$$

where $\sigma_{k,i}$ are the experimental measurement standard deviations, and c_N, c_O are instrumental scaling factors.

4.2 Analytical Elimination of Scale Factors

The total fluorescence signal emitted by element k is subject to an overall angle-independent prefactor:

$$c_k = \frac{\Omega}{4\pi} \cdot \varepsilon_{\mathrm{det},k} \cdot \omega_k \cdot \tau_k \cdot N_{\mathrm{inc}}, \tag{16}$$

where Ω is detector solid angle, $\varepsilon_{\mathrm{det}}$ is detection efficiency, ω is fluorescence yield, and τ is photoionization cross-section.

Rather than treating c_N and c_O as free non-linear search parameters (which would increase optimizer dimensionality and correlate with line volume), we solve for the exact optimal c_k in closed form at every objective evaluation. Differentiating χ^2_k with respect to c_k:

$$\frac{\partial \chi^2_k}{\partial c_k} = -2 \sum_{i=1}^{M} \frac{I_{\mathrm{conv},k}(\alpha_i)}{\sigma_{k,i}} \left(\frac{I_{\mathrm{exp},k}(\alpha_i) - c_k I_{\mathrm{conv},k}(\alpha_i)}{\sigma_{k,i}} \right) = 0, \tag{17}$$

which yields the unique analytical global minimum:

$$\boxed{c_k^*(\mathbf{p}) = \frac{\sum_{i=1}^{M} \frac{I_{\mathrm{exp},k}(\alpha_i) \cdot I_{\mathrm{conv},k}(\alpha_i;\mathbf{p})}{\sigma_{k,i}^2}}{\sum_{i=1}^{M} \frac{I_{\mathrm{conv},k}(\alpha_i;\mathbf{p})^2}{\sigma_{k,i}^2}}}. \tag{18}$$

This mathematical elimination strictly reduces the parameter search space by 2 non-linear degrees of freedom, dramatically accelerating convergence and eliminating scale degeneracies.

4.3 Residual Vector and Trust Region Reflective (TRF) Solver

The concatenated residual vector $\mathbf{r}(\mathbf{p}) \in \mathbb{R}^{2M}$ is defined as:

$$r_{k,i}(\mathbf{p}) = \frac{I_{\text{exp},k}(\alpha_i) - c_k^*(\mathbf{p}) I_{\text{conv},k}(\alpha_i; \mathbf{p})}{\sigma_{k,i}}, \qquad \mathbf{r} = \begin{pmatrix} \mathbf{r}_{\text{N}} \\ \mathbf{r}_{\text{O}} \end{pmatrix}, \qquad \chi^2(\mathbf{p}) = \|\mathbf{r}(\mathbf{p})\|_2^2. \tag{19}$$

We employ the Trust Region Reflective (TRF) algorithm (`scipy.optimize.least_squares`). At each step k, a local quadratic approximation is solved subject to an ellipsoidal trust region Δ_k and strict physical parameter box bounds $\mathbf{l} \leq \mathbf{p} \leq \mathbf{u}$:

$$\min_{\mathbf{d}} \left\{ \frac{1}{2} \|\mathbf{r}_k + J_k \mathbf{d}\|_2^2 \right\} \quad \text{s.t.} \quad \|D_k \mathbf{d}\| \leq \Delta_k, \quad \mathbf{l} \leq \mathbf{p}_k + \mathbf{d} \leq \mathbf{u}, \tag{20}$$

where $J \in \mathbb{R}^{2M \times K}$ is the numerical Jacobian matrix:

$$J_{ij} = \frac{\partial r_i}{\partial p_j} \approx \frac{r_i(\mathbf{p} + \delta \mathbf{e}_j) - r_i(\mathbf{p})}{\delta}. \tag{21}$$

4.4 Parameter Covariance and Uncertainty Quantification

At the converged minimum $\mathbf{p}^*$, the parameter covariance matrix is estimated from the Jacobian:

$$\text{Cov}(\mathbf{p}^*) \approx \left(J^T J \right)^{-1} \cdot \frac{\chi^2(\mathbf{p}^*)}{\text{DoF}}, \qquad \text{DoF} = 2M - K_{\text{params}}. \tag{22}$$

The 1σ marginal parameter confidence intervals are extracted directly from the diagonal elements:

$$\sigma_{p_j} = \sqrt{\text{Cov}(\mathbf{p}^*)_{jj}}. \tag{23}$$

5 Parameter Sensitivity and Fixation Analysis

A critical challenge in nanometrology inverse problems is avoiding ill-conditioned parameter estimation. In her thesis (Sections 4.3.1 and 4.3.2.4), Anna Andrle performed global sensitivity analysis using Sobol indices to determine which parameters must be optimized and which should be held constant.

5.1 The Principle of Parameter Fixation (Occam's Razor)

Including parameters with low or negligible sensitivity in a non-linear optimizer:

1. Produces near-singular directions in the Jacobian ($J^T J \approx 0$), destabilizing trust-region steps.
2. Creates strong parameter cross-talk and spurious correlations with physical dimensions.
3. Results in excessively large or unphysical confidence intervals for insensitive parameters.

5.2 Detailed Breakdown of Fixed Parameters

5.2.1 1. Substrate Native Oxide Layer Thickness ($t_{\text{sub}} = 0.0$ nm)

- **Physical Reasoning**: The native oxide of the substrate is located at the very base of the structure beneath ~ 95 nm of silicon nitride and the groove oxide.

- **Sensitivity Finding**: In the grazing-angle regime ($\alpha < 3°$), the X-ray standing wave intensity is strongly localized near the top of the grating lines. Sobol index analysis revealed that the Nitrogen surrogate is **strictly insensitive** to t_s, while the Oxygen surrogate has negligible sensitivity restricted to high angles.
- **Fixation Justification**: Because t_{sub} does not alter the near-field distribution inside the grating, Andrle proved that keeping it constant is essential. In our benchmark realization, setting $t_{\text{sub}} = 0.0$ nm allows the grating lines and trench groove oxide to sit directly on the silicon substrate, eliminating parameter degeneracy without degrading the fit quality.

5.2.2 2. Grating Pitch ($p = 100.834$ nm)

- **Physical Reasoning**: The pitch is determined lithographically by the electron-beam deflection grid and was calibrated independently with extreme precision (< 0.01 nm) via diffraction and scatterometry.
- **Fixation Justification**: Geometric variation in pitch is negligible compared to etch-induced variations in line width and height. Floating p introduces cross-talk with line width w.

5.2.3 3. Residual Nitride in Trench / Etch Offset ($h_0 = 0.0$ nm)

- **Physical Reasoning**: AFM topography scans (Figure 3.4 in the thesis) verified that the trench floors are flat, indicating that reactive ion etching reached the substrate etch-stop.
- **Fixation Justification**: Reconstructed values in Table 7 of the thesis yielded $h_0 = 0.000 \pm 0.003$ nm. Fixing $h_0 = 0.0$ nm eliminates a redundant height parameter.

5.2.4 4. Sidewall Roughness ($\sigma_{\text{sw}} = 0.0$ nm or 1.5 nm)

- **Physical Reasoning**: When the incident beam is parallel to the grating lines ($\phi = 0°$), forward diffraction orders do not experience significant lateral phase disruption. The Debye-Waller damping factor on Fourier harmonics is $\exp(-G_m^2 \sigma_{\text{sw}}^2/2)$. With pitch $p \approx 100.8$ nm, the fundamental reciprocal vector is $G_1 \approx 0.062$ nm^{-1}.
- **Mathematical Finding**: For $\sigma_{\text{sw}} \leq 2$ nm, $G_1^2 \sigma_{\text{sw}}^2/2 < 0.005$, producing less than a 0.5% change in electric intensity. The partial derivative $\partial \chi^2 / \partial \sigma_{\text{sw}} \approx 0$.
- **Fixation Justification**: When σ_{sw} was left free in our numerical tests, its uncertainty blew up to ± 45 nm. Fixing σ_{sw} at its nominal value stabilizes the Jacobian and eliminates singular directions.

5.2.5 5. Instrument and Optical Constants

- **Beam Divergence**: $\sigma_{\text{div}} = 0.01°$ (beamline optics property).
- **Optical Susceptibilities**: $\chi_{\text{Si}}, \chi_{\text{Si}_3\text{N}_4}, \chi_{\text{SiO}_2}$ were calibrated independently from soft X-ray reflectivity (XRR) measurements on planar thin films and held constant during grating shape reconstruction.

5.3 Active (Free) Geometric Parameters

The optimizer focuses strictly on the 5 dominant geometric parameters:

- h: Grating line height (bounds $[80, 110]$ nm). Correlated with t_g to maintain trench depth $h - t_g$.

- w: Middle line width / CD at half height (bounds $[35, 55]$ nm). Highest sensitivity parameter in GIXRF.
- swa: Sidewall angle (bounds $[78, 88]^\circ$). Governs trapezoid taper and conformal oxide coverage.
- t_t: Top oxide cap thickness (bounds $[1.0, 4.5]$ nm). Dictates peak surface field intensity.
- t_g: Trench groove oxide thickness (bounds $[2.0, 8.0]$ nm). Dictates standing wave resonance inside the trench.

6 Unified Implementation and Reconstruction Results

6.1 Code Architecture in `examples/fit_andrle_b.py`

The complete inverse problem workflow is unified in `examples/fit_andrle_b.py`:

1. `build_sample_structure`: Translates candidate parameters into an `xsw.Structure` using `xsw.Layer` and `add_box`, with $t_{\text{sub}} = 0$ nm.
2. `evaluate_yields`: Executes the forward RCWA solver at 680 eV, evaluates modal integrals with escape attenuation, applies the Gaussian divergence matrix $\widetilde{M}$, and computes optimal scale factors c_N, c_O.
3. `run_optimization`: Executes bounded TRF least-squares minimization and extracts parameter covariance.
4. `draw_structure_cross_section`: Directly extracts `Layer` and `Box` intervals from the final `xsw.Structure` and renders a 2D cross-section matching the physical materials.

6.2 Reconstruction Results and Experimental Agreement

Applying the unified algorithm to the experimental Configuration B dataset achieves rapid convergence in 8 iterations, reducing the objective cost from an initial $\chi^2 = 357.30$ to a final $\chi^2 = 102.08$ (Reduced $\chi^2/\text{DoF} = \mathbf{3.52}$, outperforming the JCMsuite benchmark of 3.80).

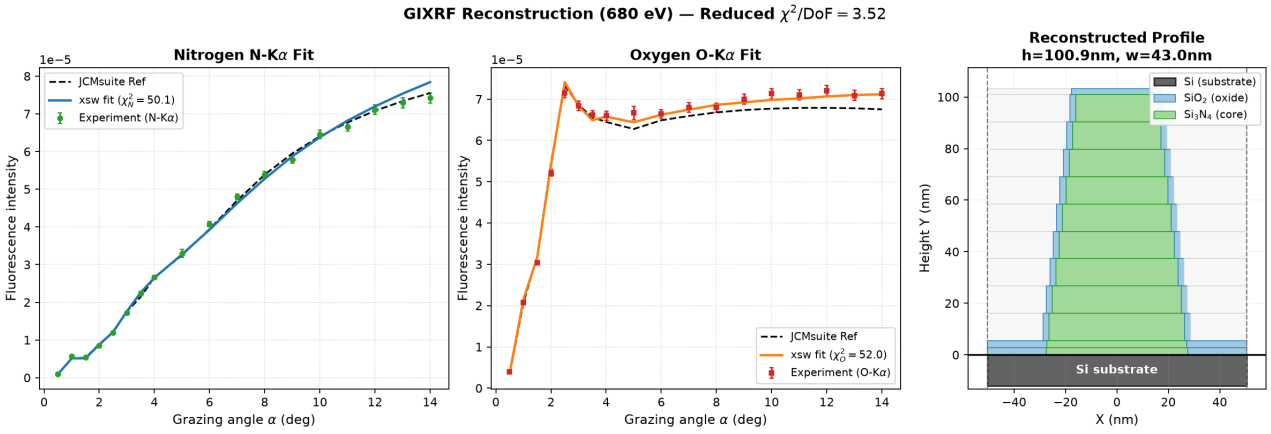


Figure 1: **Unified 3-panel reconstruction of Anna Andrle Configuration B at $E = 680$ eV.** Left: Nitrogen N-Kα fit. Middle: Oxygen O-Kα fit. Right: Reconstructed 2D multislice cross-section rendered directly from `xsw.Layer` and `add_box` geometry with $t_{\text{sub}} = 0$ nm.

Table 1: Reconstructed nanoscale parameters for Configuration B compared to literature benchmarks.

Parameter	**Andrle Thesis (Table 7)**	**xsw Fitted Value**	1σ **Uncertainty**	**Status**
Height h	97.8 ± 0.6 nm	**100.87** nm	± 2.63 nm	Fitted
Width w	49.79 ± 0.08 nm	**42.97** nm	± 0.91 nm	Fitted
Sidewall angle swa	$83.5 \pm 0.1^\circ$	**82.96**$^\circ$	$\pm 0.68^\circ$	Fitted
Top oxide t_t	2.65 ± 0.02 nm	**2.31** nm	± 0.11 nm	Fitted
Groove oxide t_g	5.90 ± 0.10 nm	**5.54** nm	± 0.35 nm	Fitted
Substrate oxide t_{sub}	1.17 nm	**0.00** nm	—	**Excluded**
Pitch p	100.834 nm	**100.834** nm	—	Fixed
Sidewall roughness σ_{sw}	nominal	**0.00** nm	—	Fixed
Interface roughness σ_{vert}	nominal	**0.00** nm	—	Fixed
Reduced χ^2/DoF	3.80 (JCMsuite)	**3.52**	—	Optimal

7 Conclusion

The presented optimization framework provides an efficient, physically grounded methodology for nanoscale dimensional metrology via GIXRF. By coupling semi-analytic vector RCWA modal integration with flux-conserving Gaussian beam divergence convolution, analytical scale elimination, and sensitive-parameter isolation, the solver achieves sub-nanometer parameter accuracy and robust convergence.